

recorded in blocks. These blocks are then linked together to create a record that is both secure and transparent for all the stakeholders who need to trace the path of a product.

How the operational framework works

Blockchain uses an organised approach to facilitate reliable, verified transaction processing throughout a decentralised system. Multiple computers work as a group to confirm all transactions and to maintain consistency and trust. There is no central managing authority involved.

Transaction flow: A transaction is initiated by a user and broadcasted to the blockchain network. This transaction contains relevant data such as sender, receiver, and amount or information being transferred.

Validation process: Network nodes verify the transaction using predefined rules and consensus mechanisms. Only valid transactions are approved and prepared for addition to the blockchain.

Block creation: Verified transactions are grouped into a block. This block is then assigned a cryptographic hash and linked to the previous block, ensuring continuity and security.

Network nodes: Nodes are responsible for validating, maintaining, and updating the blockchain. Each node holds a copy of the ledger, ensuring decentralisation and reducing the risk of data manipulation.

Consensus mechanism: Blockchain utilises consensus approaches like Proof of Work (PoW) or Proof of Stake (PoS) to maintain agreement among the participants on the entire network.

Classifications of blockchain

Blockchain networks can be classified according to their access rights and how you are allowed to use them. There are different types of blockchain networks depending on your need for the security, transparency, and scalability of the network.

Public blockchain: This is an open network where anyone can participate, view transactions, and validate data. It is fully decentralised and transparent but may be slower. Examples are Bitcoin and Ethereum.

Private blockchain: In this blockchain, only a specific enterprise controls who has access to the private network and can validate transactions. This decreases access time and increases security for the enterprise's internal business operations. Such a blockchain is used to manage internal data in an enterprise.

Consortium blockchain: Here, a group of organisations manages a semi-decentralised network instead of an individual authority. The semi-decentralised network provides the advantages of both decentralisation and controlled access, making it the perfect fit for industries where secure data needs to be accessed or shared, such as banking and supply chain. For example, multiple banks can share their banking networks with one another.

Hybrid blockchain: This combines public and private blockchains. Enterprises can store sensitive data privately while taking advantage of public transparency for certain information when necessary. This blockchain is useful for a system that contains some public data and some data that cannot be viewed by anyone externally.

Blockchain technology is being used in various sectors today including supply chain management and BFSI, where fast and secure transactions are needed, as well as in healthcare, where blockchain can protect a patient's private health information and enable secure data sharing between authorised users.

Blockchain technology continues to evolve traditional systems into advanced, reliable, digital-based solutions.

The pros and cons

The blockchain is highly secure due to the use of cryptographic techniques and has greater transparency than other systems because data is stored in an open way. As it operates without intermediaries, overall costs are reduced. It also creates a greater level of trust between all parties involved and allows for automation using smart contracts.

On the flip side, there are challenges involving its scalability, high implementation costs, and technical complexity. The lack of public awareness and regulatory issues are hindering its adoption in certain markets.

Blockchain is revolutionising the way digital systems operate today. Open source software development underwrites the rapid development of blockchains by allowing global collaboration among developers, while providing cost-efficient options for various solutions. This creates a platform that allows use by any organisation irrespective of size, in a flexible and scalable manner.

However, the challenges associated with its implementation, including scalability, complexity and regulatory issues, must be addressed to enable its widespread use. **END** 

References

- <https://ethereum.org>
- <https://www.hyperledger.org>
- <https://www.ibm.com/blockchain>
- <https://aws.amazon.com/blockchain>
- <https://azure.microsoft.com>

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